

Question 2

Elevator Simulation

Date: _____

1) PFC Chart

* Given Data

- Starts at ground floor (Floor 0)
- ~~Comparison b/w floors~~

* Processing Required

- INPUT floor requests
- Comparison b/w floor
- Update current floor variable

* Required Results.

- Print "moving up", "moving down", "doors opening" upon each successful comparison.

* Solution alternative.

- ~~INPUT~~

2) IPO chart

INPUT

- Floor requests
- No. of floor requests

Process

- Req floor > current floor
- Req floor < current floor
- Req floor = current floor.

OUTPUT

- "Moving UP"
- "Moving Down"
- "Doors opening"

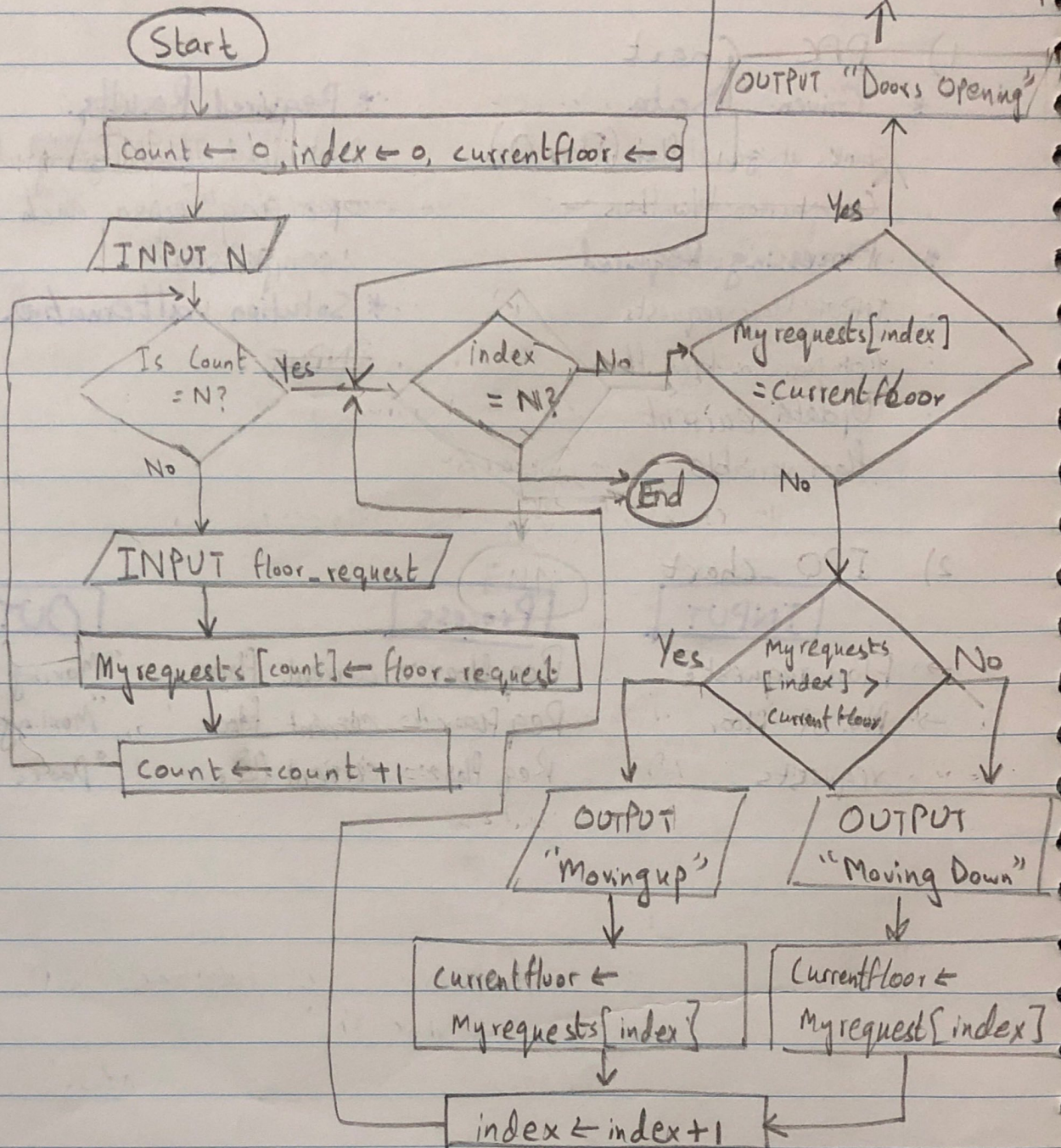
3) Flowchart

Date: _____

$c = 3$
 $N = 3$
 $r = 1, 5, 3$
 $i = 0$
 Myrequests

1	0
5	1
3	2

$1 > 0$



4) Pseudocode

Date: _____

```

DECLARE Myrequests : ARRAY OF INTEGER
DECLARE N, floor_request : INTEGER

```

```

count ← 0, index ← 0, current floor ← 0

```

```

OUTPUT "Please enter number of floor requests"

```

```

INPUT N // Number of floor requests

```

N = 5

```

FOR count 0 TO N-1

```

```

    OUTPUT "Enter floor number"

```

```

    INPUT floor_request

```

```

    Myrequest[count] ← floor_request

```

```

NEXT count

```

3	0
5	1
2	2
1	3
4	4

```

FOR index 0 TO N-1

```

```

    IF Myrequests[index] = current floor THEN

```

```

        OUTPUT "Door opening"

```

```

    NEXT index

```

C.f = 3

```

ELSE

```

```

    IF

```

```

        Myrequests[index] > current floor THEN

```

```

            OUTPUT "Moving Up"

```

```

            current floor ← Myrequests[index]

```

```

    ELSE

```

```

        OUTPUT "Moving Down"

```

```

        current floor ← Myrequest [index]

```

```

    END IF

```

```

NEXT index

```